

Week 3 Project Report

Project Title: Shopify Stock Performance Dashboard

◆ Introduction

In this project, I built a dashboard in Power BI to study **Shopify stock performance**. The goal was to show the latest values, daily changes, yearly growth, and trading activity in a clear and interactive way.

◆ Step 1: Data Preparation

- Imported the **Shopify stock dataset** (Date, Open, Close, High, Low, Volume).
- Created a **Date Table** using CALENDAR().
- Marked the Date Table as the official date table.
- Linked **Date Table[Date]** with **SHOP[date]**.

👉 This setup allowed me to use time intelligence functions like daily return and year-over-year comparison.

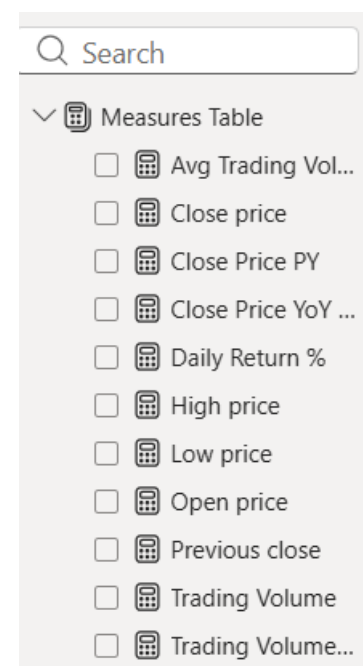
Date	Open	Close	High	Low	Volume
2023-01-02	2.78999991	2.87000009	2.91000009	2.80000009	2.78999991
2023-01-03	2.87000009	2.91000009	2.95000009	2.87000009	2.87000009
2023-01-04	2.91000009	2.95000009	2.99000009	2.91000009	2.91000009
2023-01-05	2.95000009	2.99000009	3.03000009	2.95000009	2.95000009
2023-01-06	2.99000009	3.03000009	3.07000009	2.99000009	2.99000009
2023-01-09	3.03000009	3.07000009	3.11000009	3.03000009	3.03000009
2023-01-10	3.07000009	3.11000009	3.15000009	3.07000009	3.07000009
2023-01-11	3.11000009	3.15000009	3.19000009	3.11000009	3.11000009
2023-01-12	3.15000009	3.19000009	3.23000009	3.15000009	3.15000009
2023-01-13	3.19000009	3.23000009	3.27000009	3.19000009	3.19000009
2023-01-16	3.23000009	3.27000009	3.31000009	3.23000009	3.23000009
2023-01-17	3.27000009	3.31000009	3.35000009	3.27000009	3.27000009
2023-01-18	3.31000009	3.35000009	3.39000009	3.31000009	3.31000009
2023-01-19	3.35000009	3.39000009	3.43000009	3.35000009	3.35000009
2023-01-20	3.39000009	3.43000009	3.47000009	3.39000009	3.39000009
2023-01-23	3.43000009	3.47000009	3.51000009	3.43000009	3.43000009
2023-01-24	3.47000009	3.51000009	3.55000009	3.47000009	3.47000009
2023-01-25	3.51000009	3.55000009	3.59000009	3.51000009	3.51000009
2023-01-26	3.55000009	3.59000009	3.63000009	3.55000009	3.55000009
2023-01-27	3.59000009	3.63000009	3.67000009	3.59000009	3.59000009
2023-01-30	3.63000009	3.67000009	3.71000009	3.63000009	3.63000009
2023-01-31	3.67000009	3.71000009	3.75000009	3.67000009	3.67000009
2023-02-01	3.71000009	3.75000009	3.79000009	3.71000009	3.71000009
2023-02-02	3.75000009	3.79000009	3.83000009	3.75000009	3.75000009
2023-02-03	3.79000009	3.83000009	3.87000009	3.79000009	3.79000009
2023-02-06	3.83000009	3.87000009	3.91000009	3.83000009	3.83000009
2023-02-07	3.87000009	3.91000009	3.95000009	3.87000009	3.87000009
2023-02-08	3.91000009	3.95000009	3.99000009	3.91000009	3.91000009
2023-02-09	3.95000009	3.99000009	4.03000009	3.95000009	3.95000009
2023-02-10	3.99000009	4.03000009	4.07000009	3.99000009	3.99000009
2023-02-13	4.03000009	4.07000009	4.11000009	4.03000009	4.03000009
2023-02-14	4.07000009	4.11000009	4.15000009	4.07000009	4.07000009
2023-02-15	4.11000009	4.15000009	4.19000009	4.11000009	4.11000009
2023-02-16	4.15000009	4.19000009	4.23000009	4.15000009	4.15000009
2023-02-17	4.19000009	4.23000009	4.27000009	4.19000009	4.19000009
2023-02-20	4.23000009	4.27000009	4.31000009	4.23000009	4.23000009
2023-02-21	4.27000009	4.31000009	4.35000009	4.27000009	4.27000009
2023-02-22	4.31000009	4.35000009	4.39000009	4.31000009	4.31000009
2023-02-23	4.35000009	4.39000009	4.43000009	4.35000009	4.35000009
2023-02-24	4.39000009	4.43000009	4.47000009	4.39000009	4.39000009
2023-02-27	4.43000009	4.47000009	4.51000009	4.43000009	4.43000009
2023-02-28	4.47000009	4.51000009	4.55000009	4.47000009	4.47000009
2023-02-29	4.51000009	4.55000009	4.59000009	4.51000009	4.51000009



◆ Step 2: Measures

I wrote DAX formulas to calculate important values:

- Avg Trading Volume = `AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume])`
- Close price = `AVERAGE('SHOP_2015-05-21_2025-03-16'[close])`
- Close Price PY = `CALCULATE([Close price], SAMEPERIODLASTYEAR('Date Table'[Date]))`
- Close Price YoY % = `DIVIDE([Close price] - [Close Price PY], [Close Price PY], 0)`
- Daily Return % =
- `VAR LatestDate = LASTDATE('Date Table'[Date])`
- `VAR CurrentClose =`
- `CALCULATE([Close Price], 'Date Table'[Date] = LatestDate)`
- `VAR PrevClose =`
- `CALCULATE([Close Price], DATEADD('Date Table'[Date], -1, DAY))`
- `RETURN`
- `DIVIDE(CurrentClose - PrevClose, PrevClose)`
-
- High price = `MAX('SHOP_2015-05-21_2025-03-16'[High])`
- Low price = `MIN('SHOP_2015-05-21_2025-03-16'[low])`



- Open price = `AVERAGE('SHOP_2015-05-21_2025-03-16'[Open])`
- Previous close = `CALCULATE([Close price],DATEADD('Date Table'[Date], -1, DAY))`
- Trading Volume = `SUM('SHOP_2015-05-21_2025-03-16'[volume])`
- Trading Volume YTD = `TOTALYTD([Trading Volume], 'Date Table'[Date])`

👉 These measures gave me the numbers for my visuals.

◆ Step 3: KPI Cards (Top Row)

I placed **four cards** at the top row:

1. Close Price
2. Daily Return % (latest)
3. Close Price YoY %
4. Avg Trading Volume

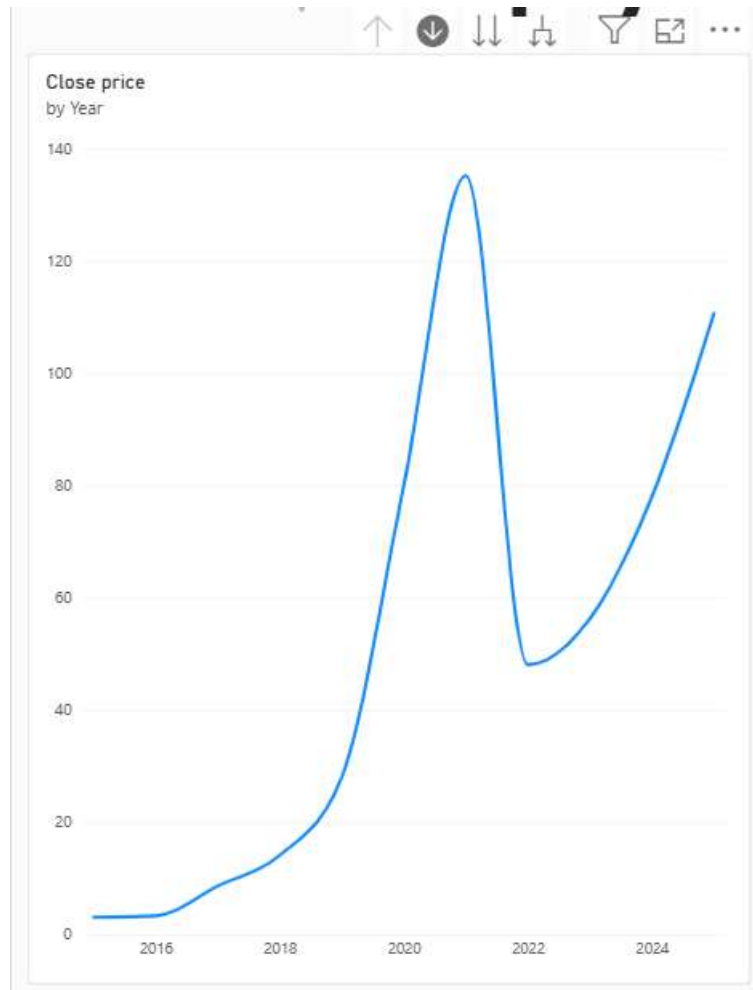
👉 These cards give a quick snapshot of the stock's performance.



◆ Step 4: Line Chart (Middle)

- Added a **Line Chart** with Date hierarchy on the X-axis and Close Price on the Y-axis.
- Enabled **drilldown and drillup** (Year → Quarter → Month → Day).

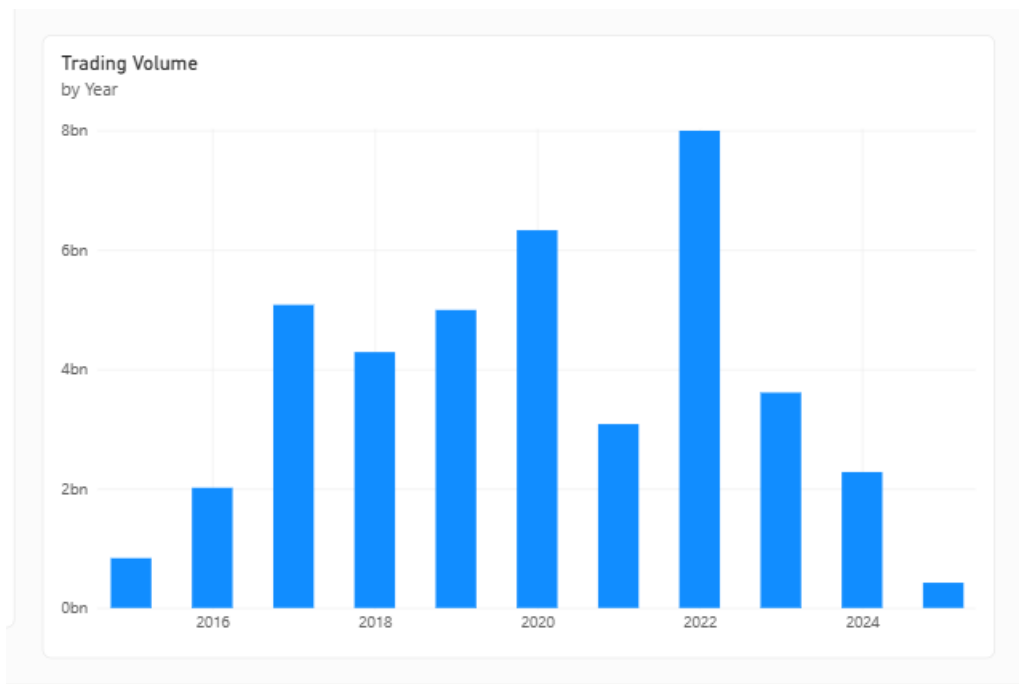
👉 This chart shows how the stock price changes over time and lets me zoom into details.



◆ Step 5: Trading Volume Chart (Bottom)

- Added a **Clustered Column Chart** with Date on the X-axis and Trading Volume on the Y-axis.
- Enabled **drilldown and drillup** here too, so I can see yearly, monthly, or daily trading activity.

👉 This chart shows how much trading happened and supports the price trend analysis.



◆ Step 6: Slicer (Right Side)

- Inserted a **Date slicer** on the right side.
- Title → *Select Date Range*.
- Format → Between slider.

👉 This slicer filters the entire dashboard based on the chosen time period.

Date, Year, Month, Day

✓ ☐ 21-05-2015 ...

✓ ☐ 22-05-2015 ...

✓ ☐ 23-05-2015 ...

✓ ☐ 24-05-2015 ...

✓ ☐ 25-05-2015 ...

✓ ☐ 26-05-2015 ...

✓ ☐ 27-05-2015 ...

✓ ☐ 28-05-2015 ...

✓ ☐ 29-05-2015 ...

✓ ☐ 30-05-2015 ...

✓ ☐ 31-05-2015 ...

✓ ☐ 01-06-2015 ...

✓ ☐ 02-06-2015 ...

✓ ☐ 03-06-2015 ...

✓ ☐ 04-06-2015 ...

✓ ☐ 05-06-2015 ...

